



Department of Information Technology

Academic Year 2024 - 2025 (Even Semester)

Degree, Semester & Branch: IV Semester B.Tech. IT

Course Code & Title: CS3452 & Theory of Computation

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Tractable and Intractable problems

- **Course Outcome:** CO5

- **Topic Learning Outcome:** TLO18

- **Activity Chosen:** Reflection

- **Justification:**

In unit V, Reflection is a powerful pedagogical tool that encourages students to think deeply about what they have learned, how they have learned it, and how the concept connects to real-world problems. For the topic *Tractable and Intractable Problems*, to enhance students' conceptual clarity, promote higher-order thinking, and bridge the gap between theory and real-world problem-solving.

- **Time Allotted for the Activity:** 5 Minutes

- **Details of the Implementation:**

- Briefly explain the importance of the topic Tractable and Intractable Problems.
- Introduce the reflection activity and its purpose: to encourage deep thinking and internalization of the concepts learned as shown in Figure 1 and 2.
- Revised the key concepts such as Definition of tractable problems (polynomial time), Intractable problems (exponential time or worse), P, NP, and NP-complete problems (at a high level), Examples from real-life scenarios
- After that questions are posted:
 - What is the difference between tractable and intractable problems?
 - Why is it important to classify problems based on their complexity?
 - Can you think of a real-world problem that seems intractable? How would you approach it?
 - What did you find most challenging or interesting in this topic?
- Ask students to write a short reflective paragraph (100–150 words) to address the guiding questions.
- Collect the written reflections.
- Provide brief feedback highlighting insightful responses and addressing misconceptions, if any.
- The papers were collected and corrected as shown in Figure 2

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	2	3	1	2	2	1	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO5	2	3	1	2	2	1	2	1	2	2	2
Justification for correlation	Apply basic Knowledge and fundamental s of Tractable and Intractable problems	The students will be able to identify the need for Tractable and Intractable problems of complex engineering problems	Able to design solutions for any applications using Tractable and Intractable problems	Able to analyze and apply Tractable and Intractable problems	Able to follow ethics for solving the problem	Able to implement the problem individually	Able to implement Tractable and Intractable problems.	Ability to recognize the need for Tractable and Intractable problems in real world applications	Exhibit knowledge in Tractable and Intractable problems for application development	Exhibit knowledge in Tractable and Intractable problems in TOC of IoT, Cloud Computing and Data Analytics to provide IT solutions	Exhibit knowledge in Tractable and Intractable problems in software projects to deliver a quality product.

- Images / Screenshot of the practice:

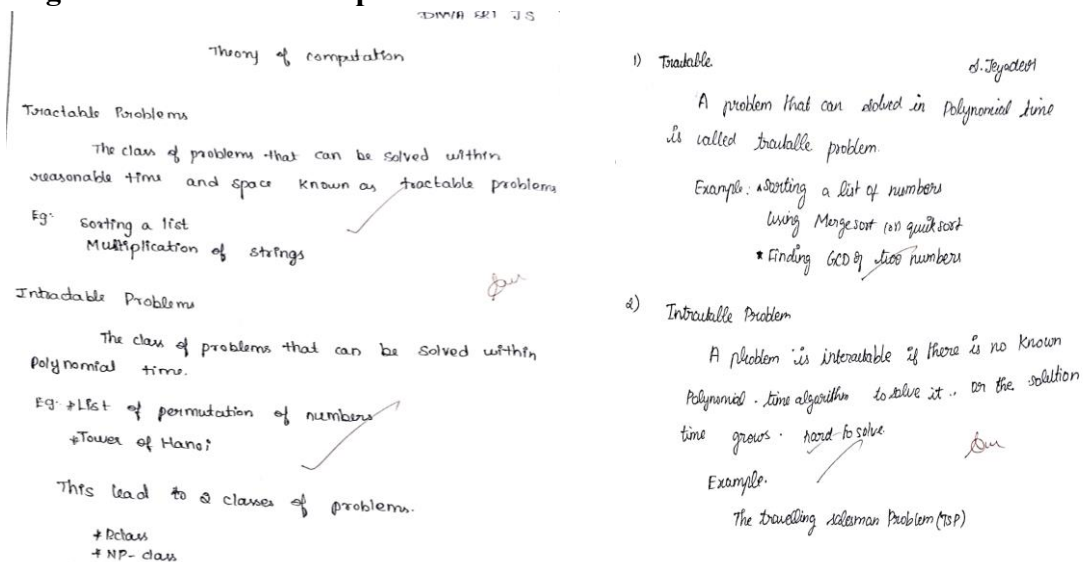


Figure 1: Reflection

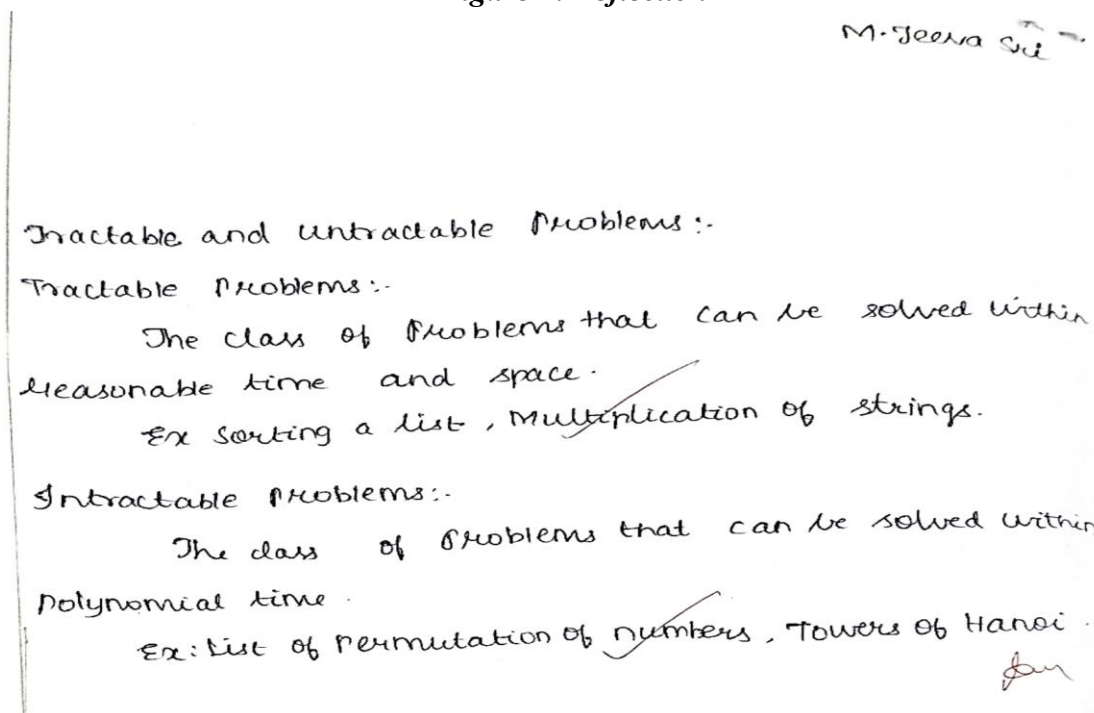


Figure 2: Student Response Ms. M. Jeeva Sri, Divya Sri J S and S. Jeya Devi

- Reflective Critique:

- ❖ Feedback of practice from students and other stakeholders:

- The reflection activity helped the students, clearly differentiate between tractable and intractable problems. Writing in their own words made the concept stick better.”
- This activity made the students to think beyond memorization. They started relating real-world problems to what we learned in class.”
- Instead of just listening, the student was able to actively participate by expressing their thoughts. It made the topic more interesting and understandable.

❖ ***Benefit of the practice:***

- From this activity the students understood how important it is in areas like cryptography and scheduling.
- Putting the understanding into words made us to feel more confident about the topic, especially before exams.
- It enhanced their understanding, encouraged higher-order thinking, and provided a platform to relate abstract concepts to real-world scenarios.
- Many students appreciated the opportunity to express their own interpretations, which strengthened their grasp of tractable and intractable problems.

❖ ***Challenges faced in implementation:***

- Some students struggled to express their understanding in their own words.
- They found it challenging to convert technical concepts into meaningful reflections without simply repeating definitions.
- A few students lacked foundational knowledge in computational complexity, which made it difficult for them to grasp the core idea of tractable and intractable problems, affecting the quality of their reflection.

References:

- ❖ <https://ohioline.osu.edu/>
- ❖ <https://www.nureva.com/blog/education/15-active-learning-activities-to-energize-your-next-college-class>
- ❖ <https://www.ritrjpm.ac.in/images/computer-science/>

Signature of Faculty Member

HOD